University of Tehran
College of Engineering
School of Mechanical Engineering

Preliminary Design and Real-System Benchmarking of Air/Helium Closed-Cycle Gas Turbine Systems

Gas Turbine Theory Project

Authors: Armin Aghazadeh
Sadra Adib

Submitted: July 2026

Abstract

This report presents a preliminary thermodynamic design and engineering comparison of closed-cycle gas turbine systems using air and helium as working fluids. The investigated cycle consists of two-stage compression with intercooling, recuperative heat exchange, external heat addition, turbine expansion, and final heat rejection. The analysis evaluates thermal efficiency, volumetric specific net work, optimum pressure ratio, recuperator effectiveness, and approximate recuperator area, mass, and cost. Air and helium are compared under the same cycle assumptions in order to identify how the working fluid changes the optimum design point and the physical size of the heat exchanger.

The project is also connected to real power-system examples. Oberhausen II is used as the closest historical benchmark because it was a plant-scale helium closed-cycle gas turbine. The HTR-PM high-temperature reactor is used as a modern helium-cooled nuclear benchmark, although its final power conversion is performed by a steam turbine rather than a direct helium gas turbine. Modern supercritical carbon dioxide closed Brayton systems are also discussed because they show that compact recuperated closed Brayton-cycle technology remains an active development path. The purpose of these comparisons is not to claim that the present cycle is identical to a current commercial plant, but to show that its component-level behavior and design trade-offs are directly connected to real external-heat power systems.

Keywords: closed-cycle gas turbine, Brayton cycle, helium, air, recuperator, pressure ratio, heat exchanger, Oberhausen II, HTR-PM, supercritical carbon dioxide.

Contents

Abstract **i**

1 Introduction **1**

1.1 Background and Motivation 1

1.2 Project Objectives 2

1.3 Report Organization 2

2 Applications and Real-System Benchmarks **3**

2.1 Closed-Cycle Gas Turbine Applications 3

2.2 Historical Benchmark: Oberhausen II 4

2.3 Modern Helium Benchmark: HTR-PM 5

2.4 Modern Closed-Brayton Benchmark: Supercritical CO_2 Systems 6

3 Cycle Model and Calculation Method **8**

3.1 Cycle Layout 8

3.2 Input Conditions and Working-Fluid Properties 9

3.3 Pressure and Temperature Relations 9

3.4 Work, Heat Input, and Efficiency 10

4 Thermal Efficiency Versus Pressure Ratio **12**

4.1 Result Description 12

5 Volumetric Specific Net Work Versus Pressure Ratio **14**

5.1 Result Description 14

6 Optimum Pressure Ratio Selection **16**

6.1 Design Criteria 16

6.2 Efficiency–Work Curves 17
6.3 State Points at the Compromise Design 18

7 Effect of Recuperator Effectiveness **20**
7.1 Effectiveness Definition 20
7.2 Air Cycle 20
7.3 Helium Cycle 21

8 Heat-Exchanger Area, Mass, and Cost **23**
8.1 NTU-Based Area Estimate 23
8.2 Helium Heat-Transfer Advantage 24
8.3 Mass and Cost Estimate 24

9 Application-Based Final Recommendation **26**
9.1 Comparison of Design Outcomes 26

10 Conclusions **29**

A MATLAB Code **30**
A.1 Running the Script 30
A.2 Structure of the Script 30
A.3 Inputs and Assumptions 30
A.4 Outputs 31
A.5 Notes on the Implementation 31
A.6 Full Listing 34

B Nomenclature **41**

List of Figures

4.1 Thermal efficiency versus total pressure ratio for air and helium. 12

5.1 Volumetric specific net work versus total pressure ratio for air and helium. 14

6.1 Air efficiency versus volumetric specific net work with selected compromise point. 17

6.2 Helium efficiency versus volumetric specific net work with selected compromise point. 17

7.1 Effect of recuperator effectiveness on air cycle efficiency. 20

7.2 Effect of recuperator effectiveness on helium cycle efficiency. 21

List of Tables

2.1 Application classification for closed-cycle gas turbine systems 4

2.2 Comparison between the present model and the Oberhausen II plant . . . 5

2.3 Comparison between the present model and HTR-PM 6

2.4 Comparison between the present model and modern sCO_2 closed Brayton systems 7

3.1 Main cycle state points 8

3.2 Working-fluid properties used in the cycle calculation 9

4.1 Maximum thermal efficiency results 13

5.1 Maximum volumetric net work results 15

6.1 Comparison of pressure-ratio selection criteria 18

6.2 Cycle state points for air at the compromise pressure ratio 18

6.3 Cycle state points for helium at the compromise pressure ratio 19

7.1 Effect of recuperator effectiveness on maximum efficiency 21

8.1 Recuperator area, mass, and cost comparison 25

9.1 Application-based working-fluid recommendations 27

A.1 Main blocks of `GTT.m` and where their results appear 31

A.2 Fixed inputs of `GTT.m` 32

A.3 Outputs of `GTT.m` 33

A.4 Main printed results of `GTT.m` 33

B.1 List of symbols used in this report 41

Chapter 1

Introduction

1.1 Background and Motivation

A closed-cycle gas turbine differs from an open-cycle gas turbine because the working fluid remains inside a sealed loop. The fluid is compressed, heated externally, expanded through a turbine, cooled, and then returned to the compressor inlet. Since the working fluid does not mix with combustion products, the cycle can use gases other than air. This is important because the thermodynamic and heat-transfer properties of the working fluid strongly influence compressor work, turbine work, pressure ratio, heat-exchanger size, and overall plant compactness.

Closed-cycle gas turbines are especially relevant for external heat sources. These heat sources may include fossil-fuel combustion through an external heater, nuclear heat, concentrated solar heat, industrial waste heat, or chemical and radioisotope heat sources for compact power systems. In such applications, the gas turbine does not need to receive combustion gases directly. This can reduce contamination of turbomachinery, allow operation with inert working fluids, and make the cycle compatible with heat sources that are not suitable for direct open-cycle combustion.

The choice between air and helium is a central design question. Air has a much higher density at the same pressure and temperature, which improves volumetric specific work and can reduce the required volume flow rate for a given power output. Helium has a much lower density but a very high specific heat and thermal conductivity. As a result, helium can improve heat-transfer performance and reduce heat-exchanger surface area, but it may increase turbomachinery volumetric-flow demands. The purpose of this project is to quantify these differences using a preliminary cycle model and then interpret the results in relation to real closed-cycle and helium-cooled systems.

1.2 Project Objectives

The main objectives of this project are:

(i) to identify practical applications of closed-cycle gas turbines and compare them with real systems;

(ii) to calculate thermal efficiency over a range of total pressure ratios for air and helium;

(iii) to calculate volumetric specific net work over the same pressure-ratio range;

(iv) to select design pressure ratios based on maximum efficiency, maximum work, and a balanced compromise point;

(v) to investigate the effect of recuperator effectiveness on cycle efficiency;

(vi) to estimate recuperator area, mass, and cost for air and helium;

(vii) to recommend suitable working fluids for different applications based on the numerical results and real-system comparisons.

1.3 Report Organization

Chapter 2 introduces closed-cycle gas turbine applications and compares the present model with real historical and modern systems. Chapter 3 presents the thermodynamic cycle model and assumptions. Chapter 4 discusses the thermal efficiency results. Chapter 5 presents the volumetric specific net work results. Chapter 6 explains the selection of optimum design points. Chapter 7 studies the effect of recuperator effectiveness. Chapter 8 estimates heat-exchanger area, mass, and cost. Chapter 9 gives the final application-based working-fluid recommendation. Chapter 10 summarizes the main conclusions.

Chapter 2

Applications and Real-System Benchmarks

2.1 Closed-Cycle Gas Turbine Applications

Closed-cycle gas turbines are suitable for applications where the heat source is external or where the working fluid must remain isolated from the environment. Historically, closed-cycle plants were considered for fossil-fuel-fired stationary power generation, nuclear power conversion, cogeneration, shipboard and underwater power units, aerospace auxiliary power, and remote power supply. Modern interest remains strongest in advanced nuclear heat-transfer systems, concentrated solar thermal power, waste heat recovery, and compact closed Brayton-cycle technologies.

Table 2.1: Application classification for closed-cycle gas turbine systems

Application	**Possible heat source**	**Main design priorities**	**Suitable working fluid**
Stationary fossil-fuel power plant	Coal, natural gas, oil, blast-furnace gas	Efficiency, reliability, fuel flexibility, simple operation	Air
High-temperature nuclear system	Nuclear reactor core	Inert heat-transfer medium, material compatibility, low neutron absorption	Helium
Solar thermal power system	Concentrated solar radiation	Effective recuperation, clean closed loop, low maintenance	Helium or air
Underwater sealed power unit	Hydrogen combustion or stored chemical heat source	Sealed operation, compactness, no atmospheric air requirement	Helium or air
Aerospace or remote power unit	Radioisotope, solar, or hydrogen heat source	Low mass, compact heat exchanger, high reliability	Helium
Modern advanced Brayton systems	Nuclear, solar, waste heat, fossil heat	Compact turbomachinery, strong recuperation, high efficiency	Helium or sCO_2

The applications in Table 2.1 show that no single working fluid is best for every case. For a conventional stationary plant, cost and simplicity are important, and air is attractive because it is available, non-expensive, and relatively dense. For nuclear and compact external-heat systems, helium becomes more attractive because it is chemically inert and has superior heat-transfer properties. This difference between volumetric work and heat-transfer performance appears repeatedly in the numerical results of this project.

2.2 Historical Benchmark: Oberhausen II

The closest real historical comparison for the helium case is the Oberhausen II helium closed-cycle gas turbine plant in Germany. The plant is important because it was not only a theoretical proposal; it represented plant-scale closed-cycle helium turbine operation. It therefore provides a useful benchmark for interpreting the present project as a preliminary version of a real engineering concept.

The present model and Oberhausen II share the same basic physical idea: a gas circulates in a closed loop, receives heat from an external source, expands through a turbine, transfers heat through internal heat exchangers, and returns to the compressor inlet. The similarity is strongest for the helium case because both systems use helium as the closed-loop working fluid. The major difference is the level of detail. Oberhausen II included real

turbomachinery, leakage control, heat-exchanger design, auxiliary systems, part-load behavior, mechanical losses, and operational constraints. The present model is instead a cycle-level design study used to identify trends and preliminary optimum points.

Table 2.2: Comparison between the present model and the Oberhausen II plant

Item	**Present project**	**Oberhausen II**
Cycle type	Closed Brayton gas turbine	Helium closed-cycle gas turbine plant
Working fluid	Air or helium	Helium
Heat source	General external heater	Real external heat source for plant operation
Heat recovery	Recuperator modeled through effectiveness	Real heat-exchanger system
Compression arrangement	Two-stage compression with intercooling	Real compressor system with plant-specific design
Design level	Preliminary thermodynamic and heat-exchanger comparison	Plant-scale engineering system
Main relevance	Shows how air and helium change efficiency, work, and heat-exchanger size	Closest historical plant-scale analogue for helium closed-cycle operation

The comparison with Oberhausen II is useful when presenting the project because it shows that helium closed-cycle gas turbines have been studied and operated at meaningful scale. The numerical results in later chapters should therefore be interpreted as a simplified preliminary cycle design, not as a purely abstract thermodynamics exercise.

2.3 Modern Helium Benchmark: HTR-PM

A modern helium-related benchmark is the HTR-PM high-temperature gas-cooled reactor at Shidao Bay in China. This plant uses helium as the reactor coolant and graphite as the moderator. Its final power conversion is performed by a steam turbine, so it is not the same as a direct helium Brayton gas turbine. However, it is still a strong modern comparison because it demonstrates the current industrial relevance of helium in high-temperature energy systems.

The connection between the present project and HTR-PM is based on helium heat transfer and high-temperature external heating. In the present project, helium is used directly as the working fluid in the closed Brayton cycle. In HTR-PM, helium transports heat from the reactor core to the steam generator. In both cases, helium is attractive because it is inert and suitable for high-temperature heat transport. The main difference is that HTR-PM uses an indirect Rankine power cycle after the helium loop, while the present project studies direct Brayton-cycle expansion of the working gas.

Table 2.3: Comparison between the present model and HTR-PM

Item	**Present project**	**HTR-PM**
Helium role	Working fluid in the power cycle	Reactor coolant and heat-transfer medium
Power conversion	Direct closed Brayton gas turbine	Steam turbine Rankine cycle
Heat source	General external heater	Nuclear reactor core
Similarity	Helium, high-temperature heat transfer, closed loop	Helium, high-temperature nuclear heat transfer, closed primary loop
Difference	Gas turbine directly expands helium or air	Helium transfers heat to steam generator
Main relevance	Shows potential of helium as a Brayton working fluid	Shows current industrial use of helium in high-temperature nuclear systems

This benchmark is important for the final interpretation of Part 6. The helium cycle does not give the highest volumetric work, but it gives a strong heat-exchanger advantage. HTR-PM shows why this heat-transfer advantage is valuable in modern high-temperature systems, even when the final plant architecture is different.

2.4 Modern Closed-Brayton Benchmark: Supercritical CO_2 Systems

A second modern benchmark is the supercritical carbon dioxide closed Brayton cycle. This is not an air or helium cycle, but it is closely related to the present project because it uses a closed-loop Brayton architecture with compression, recuperation, heat addition, expansion, and cooling. The STEP Demo project is a modern pilot-scale example of this technology direction.

The main similarity is the importance of recuperation. In both the present model and sCO_2 Brayton systems, internal heat recovery strongly affects thermal efficiency. The main difference is the working fluid. Helium has excellent thermal conductivity but low density. Supercritical CO_2 has high density near its operating region, which can reduce turbomachinery size. Therefore, the comparison helps explain why modern closed Brayton research often focuses either on helium for high-temperature nuclear heat transfer or on sCO_2 for compact power conversion.

Table 2.4: Comparison between the present model and modern sCO_2 closed Brayton systems

Item	**Present project**	**Modern sCO_2 Brayton systems**
Cycle type	Closed Brayton gas turbine	Closed Brayton power cycle
Working fluid	Air or helium	Supercritical carbon dioxide
Recuperation	Important for efficiency	Essential for high efficiency
Main fluid advantage	Air has high volumetric work; helium has strong heat transfer	High density and compact turbomachinery
Main challenge	Helium low density; air lower heat-transfer coefficient	High pressure, real-gas behavior, seals, bearings, compact heat exchangers
Main relevance	Preliminary air/helium comparison	Modern example showing continued development of recuperated closed Brayton systems

Together, Oberhausen II, HTR-PM, and sCO_2 Brayton systems provide a realistic context for the present project. Oberhausen II is the closest direct historical analogue. HTR-PM is a modern helium heat-transfer example. Supercritical CO_2 systems are modern closed-Brayton examples. The project is therefore best described as a preliminary design and benchmarking study for air/helium closed Brayton gas turbine systems for external-heat power applications.

Chapter 3

Cycle Model and Calculation Method

3.1 Cycle Layout

The analyzed system is a closed-cycle gas turbine with two-stage compression, intercooling, recuperation, external heat addition, turbine expansion, and final cooling. The main state points are defined as follows:

Table 3.1: Main cycle state points

Point	Description
1	Inlet of the low-pressure compressor
2	Outlet of the low-pressure compressor
3	Outlet of the intercooler and inlet of the high-pressure compressor
4	Outlet of the high-pressure compressor
5	Outlet of the recuperator cold side
6	Turbine inlet after external heating
7	Turbine outlet and inlet of the recuperator hot side
8	Outlet of the recuperator hot side and inlet of the final cooler

The cycle is closed, so the working fluid leaving the cooler returns to the compressor inlet. The final cooler restores the fluid to the initial compressor inlet temperature before the next cycle.

3.2 Input Conditions and Working-Fluid Properties

The specified operating conditions are

$$P_{01} = 20 \text{ bar}, \tag{3.1}$$
$$T_{01} = 300 \text{ K}, \tag{3.2}$$
$$T_{03} = 300 \text{ K}, \tag{3.3}$$
$$T_{06} = 1100 \text{ K}. \tag{3.4}$$

The component performance parameters are

$$\eta_c = 0.89, \tag{3.5}$$
$$\eta_t = 0.88, \tag{3.6}$$
$$\epsilon = 0.7. \tag{3.7}$$

The pressure losses are represented as fixed fractional pressure drops. The intercooler and cooler each have a 1% pressure loss. The recuperator has a 2.5% pressure loss on both cold and hot sides. The external heater has a 3% pressure loss.

Table 3.2: Working-fluid properties used in the cycle calculation

Fluid	c_p (kJ/kg.K)	γ	R (J/kg.K)
Air	1.005	1.400	287.0
Helium	5.193	1.666	2077.0

The constant-property approach is appropriate for a preliminary cycle comparison. It is especially accurate for helium because helium specific heat is nearly constant over a wide temperature range. For air, the constant-property approximation is less exact at high temperature, but it is acceptable for comparing the overall behavior of air and helium under the same modeling assumptions.

3.3 Pressure and Temperature Relations

The total compressor pressure ratio is defined as

$$r_c = \frac{P_{04}}{P_{01}}. \tag{3.8}$$

The compression is divided between the two compressor stages using

$$\frac{P_{02}}{P_{01}} = \sqrt{r_c}, \tag{3.9}$$

which gives

$$P_{02} = P_{01}\sqrt{r_c}, \qquad P_{04} = P_{01} r_c. \tag{3.10}$$

The intercooler pressure loss gives

$$P_{03} = 0.99 P_{02}. \tag{3.11}$$

The compressor outlet temperatures are calculated from the polytropic compression relation:

$$T_{02} = T_{01}\left(\frac{P_{02}}{P_{01}}\right)^{\frac{\gamma-1}{\gamma\eta_c}}, \tag{3.12}$$

$$T_{04} = T_{03}\left(\frac{P_{04}}{P_{03}}\right)^{\frac{\gamma-1}{\gamma\eta_c}}. \tag{3.13}$$

After the cold side of the recuperator and the external heater,

$$P_{05} = 0.975 P_{04}, \qquad P_{06} = 0.97 P_{05}. \tag{3.14}$$

The turbine exit-side pressures are calculated from the cooler and hot-side recuperator pressure losses:

$$P_{08} = \frac{P_{01}}{0.99}, \qquad P_{07} = \frac{P_{08}}{0.975}. \tag{3.15}$$

The turbine exit temperature is calculated as

$$T_{07} = T_{06}\left(\frac{P_{07}}{P_{06}}\right)^{\frac{(\gamma-1)\eta_t}{\gamma}}. \tag{3.16}$$

The recuperator cold-side outlet temperature is

$$T_{05} = T_{04} + \epsilon(T_{07} - T_{04}). \tag{3.17}$$

The hot-side recuperator outlet temperature is

$$T_{08} = T_{07} - (T_{05} - T_{04}). \tag{3.18}$$

3.4 Work, Heat Input, and Efficiency

The compressor work, turbine work, net work, and heat input are calculated as

$$w_{c1} = c_p(T_{02} - T_{01}), \tag{3.19}$$
$$w_{c2} = c_p(T_{04} - T_{03}), \tag{3.20}$$
$$w_t = c_p(T_{06} - T_{07}), \tag{3.21}$$
$$w_{net} = w_t - w_{c1} - w_{c2}, \tag{3.22}$$
$$q_{in} = c_p(T_{06} - T_{05}). \tag{3.23}$$

The thermal efficiency is

$$\eta_{th} = \frac{w_{net}}{q_{in}} \times 100. \tag{3.24}$$

The volumetric specific net work is calculated using the compressor inlet density:

$$W_{net,vol} = w_{net}\rho_1, \tag{3.25}$$

where

$$\rho_1 = \frac{P_{01}}{RT_{01}}. \tag{3.26}$$

At the specified compressor inlet condition, the densities are

$$\rho_{air} = 23.23 \text{ kg/m}^3, \tag{3.27}$$
$$\rho_{He} = 3.21 \text{ kg/m}^3. \tag{3.28}$$

This difference is central to the interpretation of the results. Air has much greater volumetric work potential because of its higher density. Helium has lower volumetric work but better thermal transport behavior.

Chapter 4

Thermal Efficiency Versus Pressure Ratio

4.1 Result Description

Figure 4.1 shows the thermal efficiency of the air and helium cycles as a function of total compressor pressure ratio. Both fluids show the same general trend: efficiency increases rapidly at low pressure ratio, reaches a maximum, and then decreases as pressure ratio continues to increase.

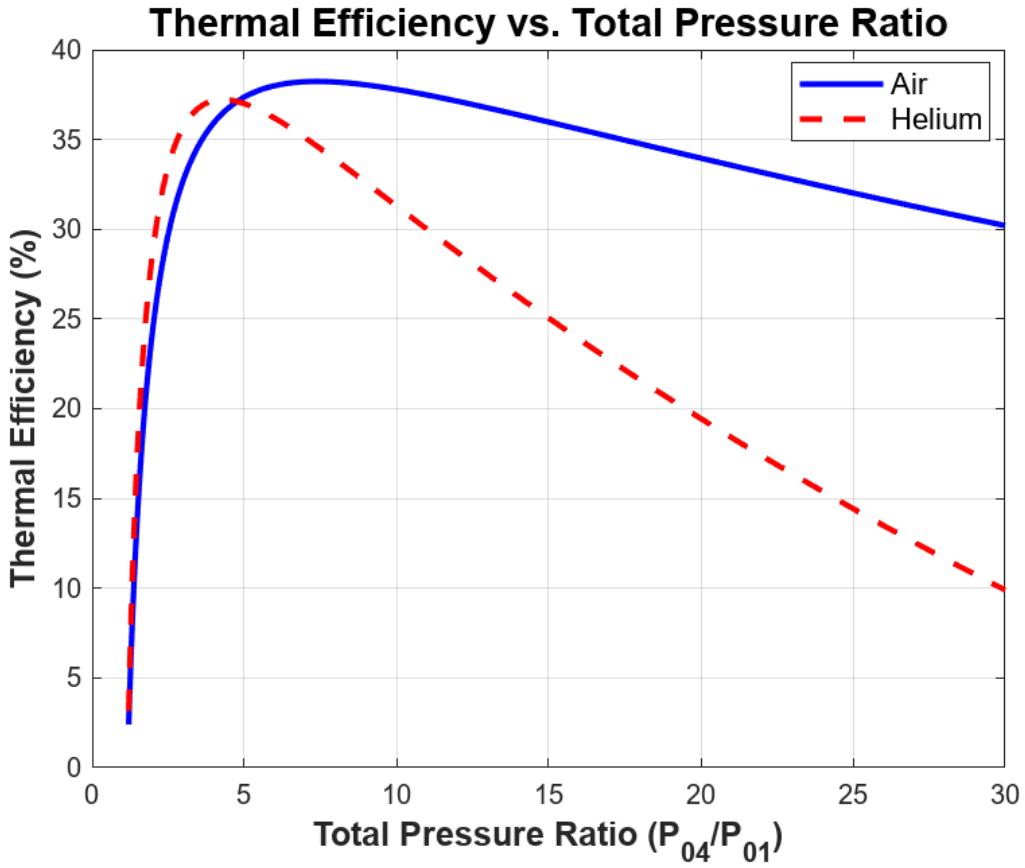


Figure 4.1: Thermal efficiency versus total pressure ratio for air and helium.

At low pressure ratio, the turbine expansion is weak, and the net work output is small. In this region, increasing pressure ratio improves the cycle because the turbine produces more useful work. The efficiency therefore rises steeply. Near the optimum point, the gain in turbine work and the reduction in required heat input are balanced by compressor work. Beyond the optimum point, the compressor work becomes too large, and the turbine exhaust temperature falls. This reduces the temperature difference available for recuperation and causes the efficiency to decrease.

Table 4.1: Maximum thermal efficiency results

Working fluid	**Pressure ratio**	**Maximum efficiency (%)**	**Net work (kJ/m^3)**
Air	7.40	38.24	4456.35
Helium	4.28	37.23	3090.81

Air reaches its maximum thermal efficiency at a pressure ratio of 7.40, while helium reaches its maximum at a lower pressure ratio of 4.28. This difference is caused mainly by the higher heat-capacity ratio of helium. Helium experiences larger temperature changes during compression and expansion for the same pressure ratio, so its optimum occurs earlier. Air can operate efficiently at a higher pressure ratio because its temperature rise during compression is less severe.

The efficiency values are close, but the curves are not identical. Air gives the slightly higher maximum efficiency in this constant-property calculation. Helium reaches its best point earlier and then loses efficiency more strongly at high pressure ratio. This result is important for design because it shows that helium should not simply be operated at the same pressure ratio as air. A helium closed-cycle gas turbine requires a lower optimum pressure ratio to avoid excessive compressor temperature rise and reduced recuperation benefit.

This behavior is consistent with real closed-cycle design practice. Historical helium closed-cycle gas turbines such as Oberhausen II were not designed by maximizing pressure ratio alone. The selected operating point had to balance thermodynamic efficiency, heat-exchanger performance, turbomachinery size, leakage control, material limits, and part-load operation. Modern high-temperature helium systems such as HTR-PM also show that high-temperature gas systems are constrained by heat-transfer and materials considerations, not only by the ideal thermodynamic optimum.

Chapter 5

Volumetric Specific Net Work Versus Pressure Ratio

5.1 Result Description

Figure 5.1 shows the volumetric specific net work as a function of total pressure ratio. This result is one of the most important comparisons between air and helium because turbomachinery size is strongly related to volume flow rate. A working fluid with high work per unit volume can produce more power for a given flow passage size.

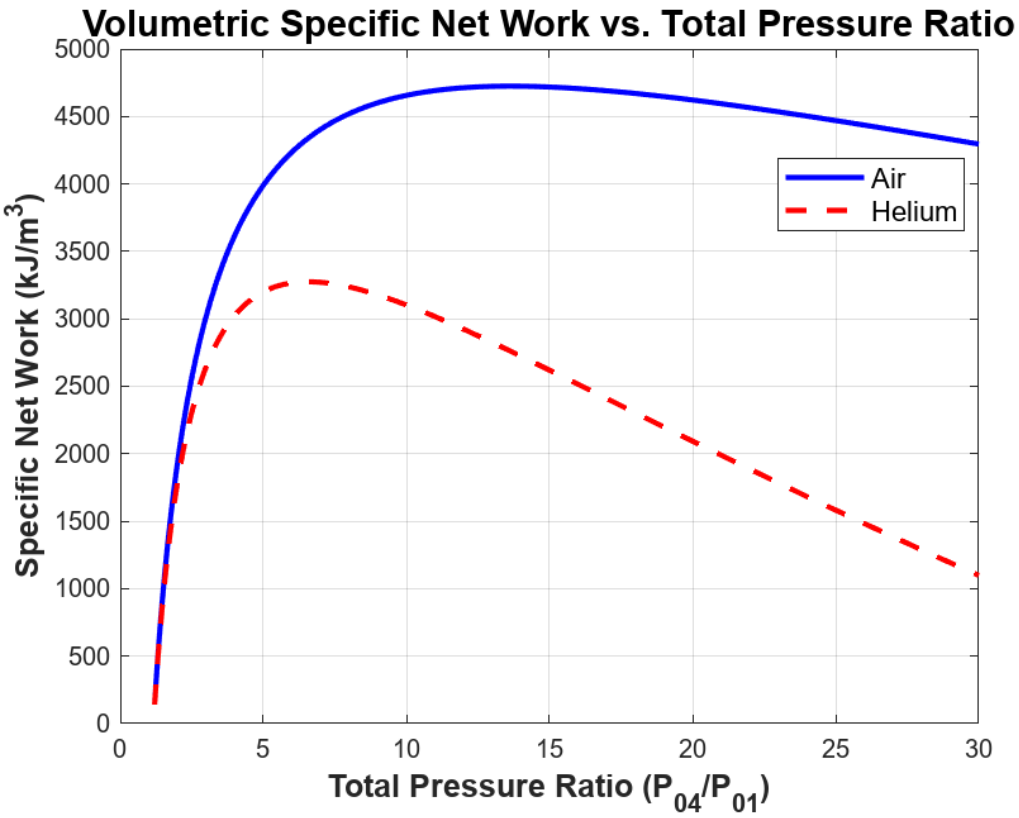


Figure 5.1: Volumetric specific net work versus total pressure ratio for air and helium.

Both curves increase at first, reach a maximum, and then decrease. At low pressure ratio, the turbine expansion is small, so the net work is low. As pressure ratio increases, turbine work increases and the net work rises. At high pressure ratio, the compressor work becomes large enough to reduce the net output, so the curve turns downward.

Table 5.1: Maximum volumetric net work results

Working fluid	**Pressure ratio**	**Efficiency (%)**	**Maximum net work (kJ/m^3)**
Air	13.63	36.53	4725.29
Helium	6.56	35.60	3274.79

The maximum-work pressure ratio is higher than the maximum-efficiency pressure ratio for both fluids. This is expected because maximum net work and maximum thermal efficiency are different design objectives. A cycle designed for maximum work accepts a lower efficiency in exchange for a higher power density. A cycle designed for maximum efficiency avoids excessive pressure ratio so that recuperation remains effective and compressor work remains moderate.

The air curve is significantly above the helium curve. This does not mean that helium is a poor working fluid in every sense. It means that helium has lower work output per unit inlet volume because its density is much lower than air at the same pressure and temperature. For the same compressor inlet condition, air has an inlet density of 23.23 kg/m^3, while helium has an inlet density of only 3.21 kg/m^3. This large density difference dominates the volumetric net work result.

This result explains why helium turbomachinery requires careful design. Helium can transfer heat effectively, but its low density increases the required volume flow rate for a given power level. Modern closed Brayton systems using supercritical CO_2 address the compactness problem in the opposite way. Instead of using a low-density gas with excellent heat transfer, they use a dense real fluid to reduce turbomachinery size. This is why STEP Demo is a useful modern comparison: it shows that closed Brayton-cycle development is strongly influenced by the trade-off between fluid density, heat transfer, and component compactness.

Chapter 6

Optimum Pressure Ratio Selection

6.1 Design Criteria

The previous two chapters show that the best pressure ratio depends on the selected objective. The pressure ratio for maximum thermal efficiency is not the same as the pressure ratio for maximum volumetric work. For this reason, three design criteria are evaluated:

(i) maximum thermal efficiency;

(ii) maximum volumetric specific net work;

(iii) a compromise point based on normalized distance from an ideal point.

The compromise method normalizes thermal efficiency and volumetric specific work:

$$W_{norm} = \frac{W_{net} - W_{net,min}}{W_{net,max} - W_{net,min}}, \tag{6.1}$$

$$\eta_{norm} = \frac{\eta - \eta_{min}}{\eta_{max} - \eta_{min}}. \tag{6.2}$$

The ideal point is defined as

$$(W_{norm}, \eta_{norm}) = (1, 1). \tag{6.3}$$

The selected compromise point is the operating point with the smallest distance from the ideal point:

$$d = \sqrt{(W_{norm} - 1)^2 + (\eta_{norm} - 1)^2}. \tag{6.4}$$

6.2 Efficiency–Work Curves

Figure 6.1 shows the air efficiency–work curve. The curve indicates that efficiency and work initially improve together. Near the top-right region, however, the curve bends. This means that further increase in work output requires sacrificing efficiency. The selected point is located close to both high efficiency and high work, so it represents a balanced preliminary design.

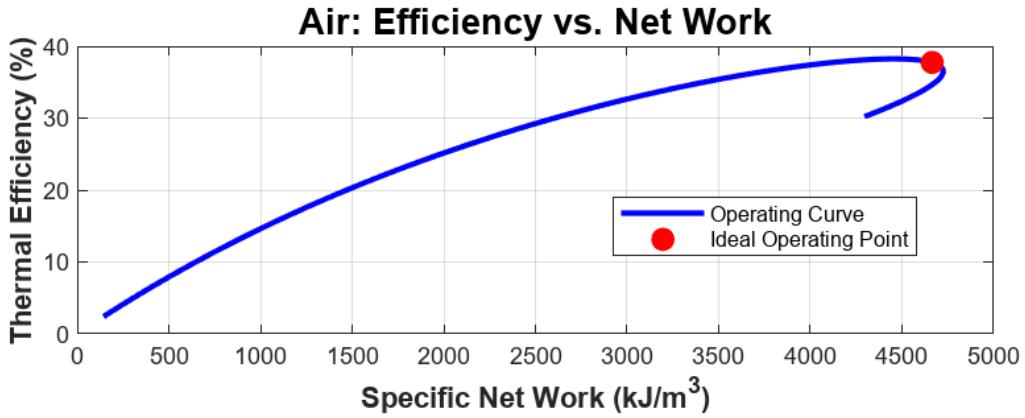


Figure 6.1: Air efficiency versus volumetric specific net work with selected compromise point.

Figure 6.2 shows the same type of curve for helium. The curve has a lower volumetric work range than air, again because helium has much lower density. The selected point is close to the best combined region of the curve. The loop-like shape reflects the fact that different pressure ratios can produce similar efficiency levels but different work outputs.

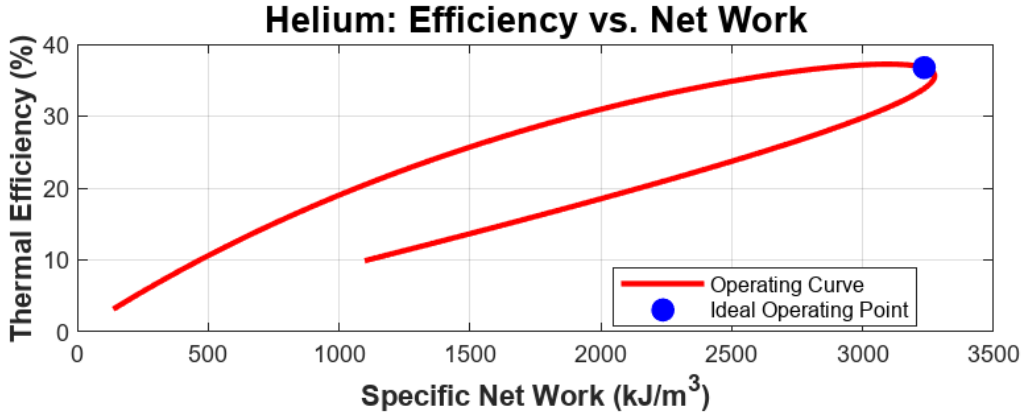


Figure 6.2: Helium efficiency versus volumetric specific net work with selected compromise point.

Table 6.1: Comparison of pressure-ratio selection criteria

Fluid	Criterion	Pressure ratio	Efficiency (%)	Net work (kJ/m^3)
Air	Maximum efficiency	7.40	38.24	4456.35
Air	Maximum work	13.63	36.53	4725.29
Air	Compromise	10.19	37.74	4665.13
Helium	Maximum efficiency	4.28	37.23	3090.81
Helium	Maximum work	6.56	35.60	3274.79
Helium	Compromise	5.38	36.74	3235.14

The compromise pressure ratio is 10.19 for air and 5.38 for helium. These values are between the maximum-efficiency and maximum-work points. This is desirable because the compromise point avoids the two extremes. A plant designed exactly at maximum efficiency may have lower power density, while a plant designed exactly at maximum work may sacrifice too much efficiency. The compromise method therefore represents a more practical preliminary design choice.

6.3 State Points at the Compromise Design

The state points at the selected compromise pressure ratios are summarized in Tables 6.2 and 6.3. These tables show how the pressure and temperature change through the two compressors, intercooler, recuperator, heater, turbine, and cooler.

Table 6.2: Cycle state points for air at the compromise pressure ratio

Point	Pressure (bar)	Temperature (K)
1	20.00	300.00
2	63.86	435.49
3	63.22	300.00
4	203.89	436.90
5	198.79	570.52
6	192.83	1100.00
7	20.72	627.79
8	20.20	494.16

Table 6.3: Cycle state points for helium at the compromise pressure ratio

Point	Pressure (bar)	Temperature (K)
1	20.00	300.00
2	46.39	437.77
3	45.93	300.00
4	107.60	439.75
5	104.91	571.80
6	101.77	1100.00
7	20.72	628.39
8	20.20	496.34

The intercooler effect is visible at point 3, where the working fluid is returned to 300 K before entering the second compressor. This reduces the work required in the second compression stage. The recuperator effect is visible between points 4 and 5. The compressed working fluid is heated before entering the external heater, which reduces external heat input. The turbine outlet temperature at point 7 remains high enough to supply this recuperator heat. Point 8 is lower than point 7 because heat has been transferred from the turbine exhaust to the compressed gas.

These state points also show why real plant design uses compromise. Pressures in the air cycle are higher at the selected pressure ratio than in the helium cycle. The helium case reaches a similar temperature pattern at lower pressure ratio. Therefore, using the same pressure ratio for both fluids would not be a fair design comparison. Real helium plants and helium-cooled systems are designed around the specific thermodynamic response of helium rather than copied directly from air-cycle practice.

Chapter 7

Effect of Recuperator Effectiveness

7.1 Effectiveness Definition

The recuperator effectiveness is defined through the cold-side outlet temperature:

$$T_{05} = T_{04} + \epsilon(T_{07} - T_{04}). \tag{7.1}$$

Increasing ϵ raises T_{05}, which means that the gas enters the external heater at a higher temperature. Since the turbine inlet temperature is fixed at 1100 K, a higher T_{05} reduces the required external heat input. The net work changes only indirectly, so the main effect of increasing recuperator effectiveness is an increase in thermal efficiency.

7.2 Air Cycle

Figure 7.1 shows the effect of recuperator effectiveness on the air cycle. Increasing effectiveness from 0.7 to 0.8 and 0.9 raises the entire efficiency curve. The improvement is strongest at lower and moderate pressure ratios because the turbine exhaust temperature is still high enough to recover useful heat.

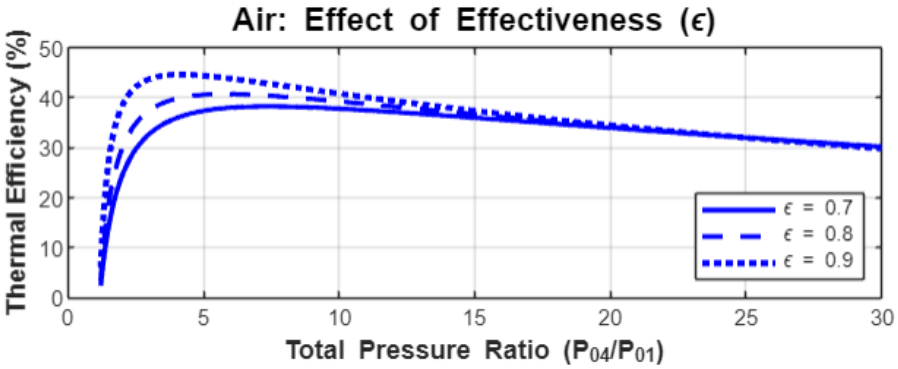


Figure 7.1: Effect of recuperator effectiveness on air cycle efficiency.

At high pressure ratios, the curves become closer. This happens because the turbine exhaust temperature decreases as the pressure ratio increases. When the exhaust temperature is lower, there is less available heat to recover in the recuperator. Therefore, even a high-effectiveness recuperator cannot improve the cycle as much at very high pressure ratio.

7.3 Helium Cycle

Figure 7.2 shows the same effect for helium. The helium curves also shift upward as effectiveness increases, but the optimum pressure ratios remain lower than the corresponding air optimum values. The curve for $\epsilon = 0.9$ reaches a higher peak and then decreases as pressure ratio increases.

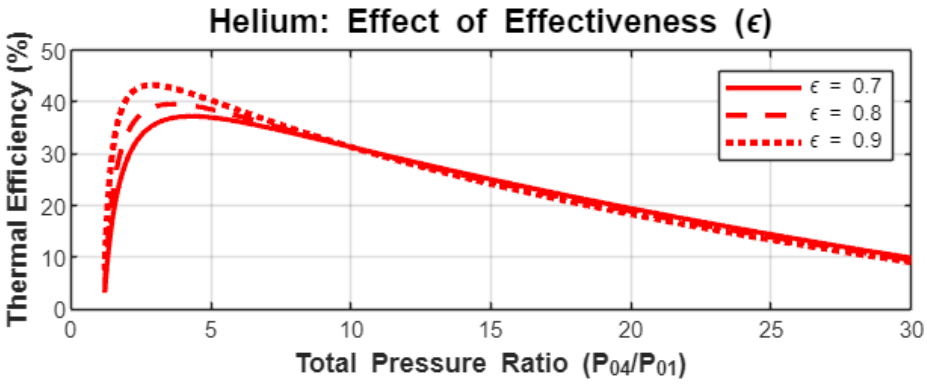


Figure 7.2: Effect of recuperator effectiveness on helium cycle efficiency.

The helium result is important because it connects directly to real high-temperature helium systems. Helium has excellent heat-transfer properties, so a compact and effective recuperator is more realistic for helium than for many other gases. This is one reason helium has been considered historically for closed-cycle gas turbines and continues to be relevant in high-temperature nuclear heat-transfer systems.

Table 7.1: Effect of recuperator effectiveness on maximum efficiency

Fluid	ϵ	Pressure ratio at maximum efficiency	Maximum efficiency (%)
Air	0.7	7.40	38.24
Air	0.8	5.78	40.71
Air	0.9	4.14	44.60
Helium	0.7	4.28	37.23
Helium	0.8	3.65	39.58
Helium	0.9	2.90	43.25

The table shows two main effects. First, higher effectiveness increases maximum efficiency for both gases. Second, the pressure ratio at maximum efficiency decreases as effectiveness increases. A highly effective recuperator favors a lower pressure ratio because the turbine

exhaust remains hot and useful for heat recovery. This trend is important for real recuperated closed Brayton systems. A designer cannot select the pressure ratio independently of the recuperator design; both must be chosen together.

Chapter 8

Heat-Exchanger Area, Mass, and Cost

8.1 NTU-Based Area Estimate

The recuperator comparison is based on the effectiveness value

$$\epsilon = 0.9. \tag{8.1}$$

For a counter-flow heat exchanger with equal heat-capacity rates, the relation between effectiveness and number of transfer units is

$$\epsilon = \frac{NTU}{1 + NTU}. \tag{8.2}$$

Therefore,

$$NTU = \frac{\epsilon}{1 - \epsilon} = 9. \tag{8.3}$$

The heat-transfer area is estimated from

$$A = \frac{NTUC_{min}}{U}. \tag{8.4}$$

A basis of

$$C_{min} = 1 \text{ kW/K} \tag{8.5}$$

is used. The assumed air-side overall heat-transfer coefficient is

$$U_{air} = 0.100 \text{ kW/(m}^2\text{.K)}. \tag{8.6}$$

The absolute heat-exchanger areas depend on this basis, but the comparison between air and helium is still meaningful because both fluids are evaluated using the same effectiveness target and heat-capacity-rate basis.

8.2 Helium Heat-Transfer Advantage

The heat-transfer comparison uses the turbulent-flow correlation

$$Nu = 0.023 Re^{0.8} Pr^{0.4}. \tag{8.7}$$

Because the Prandtl numbers of air and helium are similar, the difference in heat-transfer coefficient is mainly controlled by Reynolds number and thermal conductivity. Helium has much higher thermal conductivity than air. Although helium has lower density, its heat-transfer coefficient can still be significantly higher under comparable pressure-loss conditions.

Using the helium-to-air heat-transfer comparison from the reference closed-cycle gas turbine analysis,

$$\frac{h_{He}}{h_{air}} \approx 1.86. \tag{8.8}$$

The same ratio is applied to the overall heat-transfer coefficient:

$$U_{He} = 1.86 U_{air}. \tag{8.9}$$

This gives

$$U_{He} = 0.186 \text{ kW/(m}^2\text{.K)}. \tag{8.10}$$

8.3 Mass and Cost Estimate

The heat-exchanger mass is estimated using

$$M = \alpha A, \tag{8.11}$$

where

$$\alpha = 15 \text{ kg/m}^2. \tag{8.12}$$

The cost is estimated using

$$\log_{10}(C^0) = 4.3247 - 0.3030 \log_{10}(A) + 0.1634 [\log_{10}(A)]^2. \tag{8.13}$$

Table 8.1: Recuperator area, mass, and cost comparison

Quantity	**Air**	**Helium**	**Change using helium**
Effectiveness	0.90	0.90	Same
NTU	9.00	9.00	Same
Overall heat-transfer coefficient, U (kW/m^2K)	0.100	0.186	Higher
Area, A (m^2)	90.00	48.39	46.24% lower
Mass, M (kg)	1350.00	725.81	46.24% lower
Cost, C^0 (USD)	22730.01	18967.38	16.55% lower

The helium recuperator requires a much smaller area than the air recuperator for the same effectiveness. Since the mass model is directly proportional to surface area, the estimated mass reduction is the same as the area reduction. The cost reduction is smaller because the cost correlation is nonlinear and does not scale directly with surface area.

This is the strongest practical argument in favor of helium. Air gives better volumetric work, but helium reduces heat-exchanger size. In compact closed-cycle systems, heat exchangers can dominate the mass and volume of the plant. Therefore, the helium heat-exchanger advantage can be more important than the lower volumetric work in applications such as nuclear, aerospace, solar thermal, underwater, or remote power systems.

The result also explains the connection to modern systems. HTR-PM uses helium because helium is suitable for high-temperature nuclear heat transfer. The present cycle uses helium more directly, as the Brayton-cycle working fluid, which is closer to the historical Oberhausen II plant. Supercritical CO_2 systems take a different approach: they reduce equipment size mainly through high fluid density. All three cases show that working-fluid selection is not only a thermodynamic question; it is also a component-size and heat-transfer question.

Chapter 9

Application-Based Final Recommendation

9.1 Comparison of Design Outcomes

The final selection of working fluid depends on the application. Air gives higher volumetric specific work and is simpler to handle. Helium gives lower volumetric work but provides better heat transfer and smaller recuperator area. The pressure-ratio recommendations are summarized in Table 9.1.

Table 9.1: Application-based working-fluid recommendations

Application	**Fluid**	**Criterion**	**Pressure ratio**	**Reason**
Stationary fossil-fuel closed-cycle plant	Air	Compromise	10.19	High volumetric work, simple working fluid, and lower operating complexity
Base-load industrial power plant	Air	Work or compromise	10.19–13.63	High power density and practical cost are important
High-temperature nuclear plant	Helium	Compromise	5.38	Inert gas, low neutron absorption, and compact heat exchanger
Solar thermal closed-cycle plant	Helium	Efficiency	4.28	Efficient heat recovery and clean closed-loop operation are important
Underwater sealed power unit	Helium	Compromise	5.38	Sealed operation and compact recuperator are important
Aerospace or remote small unit	Helium	Compromise	5.38	Low heat-exchanger mass and compactness are important
Modern advanced Brayton benchmark	sCO_2 or helium	Technology-dependent	–	sCO_2 gives compact turbomachinery; helium is attractive for high-temperature nuclear heat transfer

For conventional stationary fossil-fuel systems, air is recommended. The present calculation shows that air produces significantly higher volumetric specific net work. This means that, for the same compressor inlet volume flow rate, the air cycle can produce more net output. Air is also easier to obtain, cheaper to use, and simpler from a sealing and handling point of view.

For nuclear, solar, underwater, aerospace, and remote systems, helium is recommended. In these applications, compact heat exchangers, chemical inertness, and compatibility with special heat sources are often more important than maximum volumetric work. The helium cycle also operates at a lower optimum pressure ratio, which can simplify the

compressor pressure-ratio requirement.

The overall recommendation is therefore:

> Use air when cost, simplicity, and high volumetric specific work are the main priorities.

> Use helium when compact heat exchangers, lower heat-exchanger mass, chemical inertness, and compatibility with special external heat sources are the main priorities.

Chapter 10

Conclusions

A preliminary closed-cycle gas turbine analysis was performed for air and helium. The model included two-stage compression with intercooling, recuperation, external heat addition, turbine expansion, and final cooling. The pressure ratio was varied from 1.2 to 30, and the thermal efficiency and volumetric specific net work were calculated for both fluids.

The results show that air gives higher volumetric specific net work because of its higher density. The maximum volumetric work for air is 4725.29 kJ/m^3, compared with 3274.79 kJ/m^3 for helium. Therefore, air is more suitable when power density and simpler plant operation are the main priorities.

Helium reaches its optimum efficiency at a lower pressure ratio than air. The maximum-efficiency pressure ratio is 7.40 for air and 4.28 for helium. The selected compromise pressure ratio is 10.19 for air and 5.38 for helium. This confirms that helium should not be designed using the same pressure ratio as air.

Increasing recuperator effectiveness from 0.7 to 0.9 improves efficiency for both fluids and shifts the optimum pressure ratio downward. This shows that the recuperator and pressure ratio must be selected together. The recuperator is not just an additional component; it strongly changes the optimum cycle design.

The heat-exchanger comparison gives the strongest advantage for helium. Using the estimated heat-transfer coefficient ratio, helium reduces recuperator area and mass by about 46% for the same effectiveness. This result explains why helium is attractive for compact and high-temperature systems even though it has lower volumetric specific work.

The comparison with real systems supports the engineering relevance of the project. Oberhausen II is the closest historical plant-scale example of helium closed-cycle gas turbine operation. HTR-PM is a modern helium-cooled nuclear system, although it uses steam power conversion. Modern sCO_2 systems show that recuperated closed Brayton-cycle technology remains an active development direction. Therefore, the present project is best understood as a preliminary design and benchmarking study for air/helium closed Brayton gas turbine systems for advanced external-heat power applications.

Chapter A

MATLAB Code

This appendix explains how the MATLAB script `GTT.m` produces the results in Chapters 3–8 and then reproduces the script in full (Section A.6). The script is a single file: it needs no toolboxes, no external data files and no user input. The same code is also provided as a MATLAB live script, `GT.mlx`, which stores the figures and printed results next to the code.

A.1 Running the Script

- **Requirements:** MATLAB (checked with R2023b) or GNU Octave (checked with version 8.4). Both give identical printed results.
- **How to run:** place `GTT.m` in the working folder and run it, either with the *Run* button or by typing `GTT` in the command window.
- **What it produces:** six figures (Figures 4.1, 5.1, 6.1, 6.2, 7.1 and 7.2 of this report) and a set of printed results in the command window (Tables 4.1, 5.1, 6.1–6.3 and 8.1).

A.2 Structure of the Script

Table A.1 divides the script into its main blocks. Line numbers refer to the listing in Section A.6.

A.3 Inputs and Assumptions

Table A.2 lists every fixed value in the script. The pressure-loss factors are not stored as variables; they are written directly into the pressure relations (for example `P03 = P02 * 0.99`).

Units inside the script: pressures in Pa, temperatures in K, specific work and heat in kJ/kg, gas constants in J/kg.K (`R = 8314/MW`) and densities in kg/m^3. The volumetric net work is the specific net work multiplied by the compressor-inlet density, so it is in kJ/m^3.

Table A.1: Main blocks of `GTT.m` and where their results appear

Lines	Block	What it does	Report
1–3	Initialisation	Clears the workspace and closes open figures	–
5–22	Inputs	Operating conditions, component efficiencies and gas properties	Eqs. (3.1)–(3.7), Table 3.2
23–30	Sweep set-up	1,000 total pressure ratios from 1.2 to 30; compressor-inlet densities	Eqs. (3.8), (3.26)–(3.28)
32–73	Main cycle loop	State points, work, heat input, η_{th} and $W_{net,vol}$ for air (lines 35–52) and helium (lines 54–71)	Eqs. (3.9)–(3.25)
74–92	Pressure-ratio plots	Efficiency and volumetric net work against r_c	Figures 4.1, 5.1
93–146	Maximum-efficiency point	Finds the optimum r_c and prints its state points	Tables 4.1, 6.1
147–198	Maximum-work point	Same for maximum volumetric net work	Tables 5.1, 6.1
199–273	Compromise point	Normalised trade-off between efficiency and work; plots and state points	Figures 6.1, 6.2; Tables 6.1–6.3
275–343	Recuperator effectiveness	Repeats the sweep for $\epsilon = 0.7$, 0.8 and 0.9	Figures 7.1, 7.2; Table 7.1
345–468	Heat exchanger size, mass and cost	NTU, property ratios, heat-transfer ratio, area, mass and cost at the compromise point	Eqs. (8.1)–(8.13), Table 8.1

A.4 Outputs

Table A.3 links each output of the script to the report, and Table A.4 gives the main printed values, which can be used to check a run.

A.5 Notes on the Implementation

- **Polytropic efficiencies.** The exponents $(\gamma-1)/(\gamma\eta_c)$ in lines 36 and 40 and $(\gamma-1)\eta_t/\gamma$ in line 43 make η_c and η_t polytropic efficiencies, as in Eqs. (3.12), (3.13) and (3.16).
- **Fixed turbine back pressure.** P_{07} and P_{08} are calculated once from P_{01} (lines 9–10) and are the same for every pressure ratio: 20.72 bar and 20.20 bar, Eq. (3.15).
- **Repeated cycle calculation.** The state-point equations are written out separately for air and helium and repeated for each selected design point (lines 102–123, 155–175, 231–251, 289–321 and 357–385) rather than placed in a function. All copies use the same relations, so a change to an assumption has to be made in every copy.
- **Balanced recuperator.** T_{08} follows from an energy balance with the same c_p on both

Table A.2: Fixed inputs of `GTT.m`

Variable	Symbol	Value	Meaning
`P01`	P_{01}	20 bar	Compressor inlet pressure
`T01`, `T03`	T_{01}, T_{03}	300 K	Inlet temperature of both compressors
`T06`	T_{06}	1100 K	Turbine inlet temperature
`eta_c`, `eta_t`	η_c, η_t	0.89, 0.88	Compressor and turbine polytropic efficiencies
`epsilon`	ϵ	0.7	Recuperator effectiveness, base case
`cp_air`, `gamma_air`, `MW_air`	c_p, γ, –	1.005 kJ/kg.K, 1.4, 28.97 kg/kmol	Air properties
`cp_he`, `gamma_he`, `MW_he`	c_p, γ, –	5.193 kJ/kg.K, 1.666, 4.003 kg/kmol	Helium properties
Factors 0.99, 0.975, 0.97	–	1%, 2.5%, 3%	Pressure losses: intercooler and cooler, each recuperator side, heater
`rc_range`	r_c	1.2–30, 1,000 points	Pressure-ratio sweep (step about 0.029)
`eps_opt`	ϵ	0.9	Effectiveness used to size the recuperator
`Cdot_basis`	C_{min}	1 kW/K	Sizing basis for the recuperator
`U_air`	U_{air}	0.100 kW/m^{2}K	Assumed air-side overall coefficient
`h_ratio_cohen`	h_{He}/h_{air}	1.86	Reference heat-transfer ratio [1]
`alpha`	α	15 kg/m^2	Heat-exchanger mass per unit area
In `C0`	–	4.3247, −0.3030, 0.1634	Cost-correlation coefficients, Eq. (8.13), given by the course instructor as an estimate

sides, Eq. (3.18). This is why the number of transfer units is $NTU = \epsilon/(1-\epsilon)$ in Eq. (8.3).

- **Optimum search.** The optimum pressure ratios are the maxima found on the 1,000-point grid, so they are accurate to about ±0.03.
- **Compromise (utopia) point.** Lines 202–211 rescale the efficiency and volumetric-work curves to the range 0–1 and select the pressure ratio closest to the ideal point (1, 1). This gives the compromise pressure ratios of Chapter 6: 10.19 for air and 5.38 for helium.
- **Effectiveness study.** Lines 283–323 recalculate the efficiency curves for $\epsilon = 0.7$, 0.8 and 0.9. The script plots these curves but does not print their maxima; the values in Table 7.1 are the maxima of these curves.
- **Heat-transfer ratio.** Lines 396–425 compare the two gases flowing through the same channel with the same frictional pressure drop. With the Blasius-type friction

Table A.3: Outputs of `GTT.m`

Output	**Lines**	**Report**
Figure 1: efficiency against pressure ratio	77–84	Figure 4.1
Figure 2: volumetric net work against pressure ratio	85–92	Figure 5.1
Printed maximum-efficiency points and state points	125–146	Tables 4.1, 6.1
Printed maximum-work points and state points	177–198	Tables 5.1, 6.1
Figures 3 and 4: efficiency against volumetric net work, with the compromise point	213–229	Figures 6.1, 6.2
Printed compromise points and state points	253–273	Tables 6.1–6.3
Figures 5 and 6: effect of recuperator effectiveness	325–343	Figures 7.1, 7.2; Table 7.1
Printed recuperator results	352–468	Eqs. (8.3), (8.8)–(8.10); Table 8.1

Table A.4: Main printed results of `GTT.m`

Quantity	**Air**	**Helium**
Pressure ratio at maximum efficiency	7.40	4.28
Maximum thermal efficiency (%)	38.24	37.23
Pressure ratio at maximum volumetric net work	13.63	6.56
Maximum volumetric net work (kJ/m^3)	4725.29	3274.79
Compromise pressure ratio	10.19	5.38
Thermal efficiency at the compromise point (%)	37.74	36.74
Recuperator area for $C_{min} = 1$ kW/K (m^2)	90.000	48.387
Recuperator mass (kg)	1350.000	725.806
Recuperator cost (USD)	22730.01	18967.38

law, $\Delta p \propto \rho^{0.75}\mu^{0.25}V^{1.75}$, this fixes the velocity ratio (`C_ratio`, line 415) and therefore the Reynolds-number ratio (line 416). The Dittus–Boelter correlation, $Nu = 0.023\,Re^{0.8}Pr^{0.4}$, with equal Prandtl numbers then gives (line 418)

$$\frac{h_{He}}{h_{air}} = \left(\frac{Re_{He}}{Re_{air}}\right)^{0.8} \frac{k_{He}}{k_{air}}.$$

Viscosity and conductivity are interpolated at the mean recuperator temperature, about 533 K, which lies inside the tables. The tables give μ in 10^{-7} Pa.s and k in 10^{-3} W/m.K; only ratios are used, so these scale factors cancel. The values are consistent with the gas-property tables of Incropera et al. [9].

- **Choice of 1.86.** The estimate from lines 411–418 is $h_{He}/h_{air} = 2.09$. It differs from the reference value 1.86 [1] by 12.5%, which is more than the 10% tolerance in line 421, so the script uses 1.86. This is the value in Eq. (8.8) and Table 8.1. With 2.09, the same equations would give a 52% area and mass reduction and a 19% cost reduction,

instead of 46% and 17%.

- **Absolute sizes.** Areas, masses and costs are calculated for $C_{min} = 1$ kW/K and an assumed U_{air}, so only the comparison between air and helium is meaningful. The cost correlation, Eq. (8.13), was provided by the course instructor as an estimate, so the costs are indicative.
- **Printed output.** Lines 125–126, 137, 177–178, 189, 253–255 and 264 call `fprintf` without a final newline, so some headings run into the following text in the command window (for example `max efficiencyAir`). The printed values are not affected.

A.6 Full Listing

```
clc;
clear;
close all;

P01 = 20 * 1e5;
T01 = 300;
T03 = 300;
T06 = 1100;
P08 = P01 / 0.99;
P07 = P08 / 0.975;

eta_c = 0.89;
eta_t = 0.88;
epsilon = 0.7;
cp_air = 1.005;
gamma_air = 1.4;
MW_air = 28.97;
R_air = 8314 / MW_air;
cp_he = 5.193;
gamma_he = 1.666;
MW_he = 4.003;
R_he = 8314 / MW_he;
rc_range = linspace(1.2, 30, 1000);
N = length(rc_range);
eta_th_air = zeros(1, N);
W_net_air  = zeros(1, N);
eta_th_he  = zeros(1, N);
W_net_he   = zeros(1, N);
rho1_air = P01 / (R_air * T01);
rho1_he = P01 / (R_he * T01);

for i = 1:N
    rc = rc_range(i);

    P02_air = P01 * sqrt(rc);
    T02_air = T01 * (P02_air/P01)^((gamma_air-1)/(gamma_air*eta_c));
    P03_air = P02_air * 0.99;
    T03_air = T03;
    P04_air = P01 * rc;
    T04_air = T03_air * (P04_air/P03_air)^((gamma_air-1)/(gamma_air*eta_c));
    P05_air = P04_air * 0.975;
    P06_air = P05_air * 0.97;
    T07_air = T06 * (P07/P06_air)^(((gamma_air-1)*eta_t)/gamma_air);
    T05_air = T04_air + epsilon * (T07_air - T04_air);

    wc1_air = cp_air * (T02_air - T01);
    wc2_air = cp_air * (T04_air - T03_air);
    wt_air = cp_air * (T06 - T07_air);
    wnet_mass_air = wt_air - (wc1_air + wc2_air);
    W_net_air(i) = wnet_mass_air * rho1_air;
    qin_air = cp_air * (T06 - T05_air);
    eta_th_air(i) = (wnet_mass_air / qin_air) * 100;

    P02_he = P01 * sqrt(rc);
    T02_he = T01 * (P02_he/P01)^((gamma_he-1)/(gamma_he*eta_c));
```

```
    P03_he = P02_he * 0.99;
    T03_he = T03;
    P04_he = P01 * rc;
    T04_he = T03_he * (P04_he/P03_he)^((gamma_he-1)/(gamma_he*eta_c));
    P05_he = P04_he * 0.975;
    P06_he = P05_he * 0.97;
    T07_he = T06 * (P07/P06_he)^(((gamma_he-1)*eta_t)/gamma_he);
    T05_he = T04_he + epsilon * (T07_he - T04_he);

    wc1_he = cp_he * (T02_he - T01);
    wc2_he = cp_he * (T04_he - T03_he);
    wt_he = cp_he * (T06 - T07_he);
    wnet_mass_he = wt_he - (wc1_he + wc2_he);
    W_net_he(i) = wnet_mass_he * rho1_he;
    qin_he = cp_he * (T06 - T05_he);
    eta_th_he(i) = (wnet_mass_he / qin_he) * 100;

end
%% Plotting Results (task 2 & 3)


figure;
plot(rc_range, eta_th_air, 'b-', 'LineWidth', 2); hold on;
plot(rc_range, eta_th_he, 'r--', 'LineWidth', 2);
grid on;
xlabel('Total Pressure Ratio (P_{04}/P_{01})', 'FontSize', 12, 'FontWeight', 'bold');
ylabel('Thermal Efficiency (%)', 'FontSize', 12, 'FontWeight', 'bold');
title('Thermal Efficiency vs. Total Pressure Ratio', 'FontSize', 14);
legend('Air', 'Helium', 'Location', 'best', 'FontSize', 11);
figure;
plot(rc_range, W_net_air, 'b-', 'LineWidth', 2); hold on;
plot(rc_range, W_net_he, 'r--', 'LineWidth', 2);
grid on;
xlabel('Total Pressure Ratio (P_{04}/P_{01})', 'FontSize', 12, 'FontWeight', 'bold');
ylabel('Specific Net Work (kJ/m^3)', 'FontSize', 12, 'FontWeight', 'bold');
title('Volumetric Specific Net Work vs. Total Pressure Ratio', 'FontSize', 14);
legend('Air', 'Helium', 'Location', 'best', 'FontSize', 11);
%% Optimal point(task 4)
%% Based on max efficiency


[~, idx_opt_air] = max(eta_th_air);
rc_opt_air = rc_range(idx_opt_air);
[~, idx_opt_he] = max(eta_th_he);
rc_opt_he = rc_range(idx_opt_he);

P02_a = P01 * sqrt(rc_opt_air);
T02_a = T01 * (P02_a/P01)^((gamma_air-1)/(gamma_air*eta_c));
P03_a = P02_a * 0.99;
T03_a = T03;
P04_a = P01 * rc_opt_air;
T04_a = T03_a * (P04_a/P03_a)^((gamma_air-1)/(gamma_air*eta_c));
P05_a = P04_a * 0.975;
P06_a = P05_a * 0.97;
T07_a = T06 * (P07/P06_a)^(((gamma_air-1)*eta_t)/gamma_air);
T05_a = T04_a + epsilon * (T07_a - T04_a);
T08_a = T07_a - (T05_a - T04_a);
P02_h = P01 * sqrt(rc_opt_he);
T02_h = T01 * (P02_h/P01)^((gamma_he-1)/(gamma_he*eta_c));
P03_h = P02_h * 0.99;
T03_h = T03;
P04_h = P01 * rc_opt_he;
T04_h = T03_h * (P04_h/P03_h)^((gamma_he-1)/(gamma_he*eta_c));
P05_h = P04_h * 0.975;
P06_h = P05_h * 0.97;
T07_h = T06 * (P07/P06_h)^(((gamma_he-1)*eta_t)/gamma_he);
T05_h = T04_h + epsilon * (T07_h - T04_h);
T08_h = T07_h - (T05_h - T04_h);

fprintf('Optimal point based on max efficiency');
fprintf('Air (Pressure Ratio: %.2f - Efficiency: %.2f%% - Net Work: %.2f kJ/m^3)',rc_opt_air, eta_th_air(
    idx_opt_air), W_net_air(idx_opt_air));
fprintf('Point\t Pressure (bar)\t Temperature (K)\n');
```

```
fprintf('1\t\t %5.2f\t\t %6.2f\n', P01/1e5, T01);
fprintf('2\t\t %5.2f\t\t %6.2f\n', P02_a/1e5, T02_a);
fprintf('3\t\t %5.2f\t\t %6.2f\n', P03_a/1e5, T03_a);
fprintf('4\t\t %5.2f\t\t %6.2f\n', P04_a/1e5, T04_a);
fprintf('5\t\t %5.2f\t\t %6.2f\n', P05_a/1e5, T05_a);
fprintf('6\t\t %5.2f\t\t %6.2f\n', P06_a/1e5, T06);
fprintf('7\t\t %5.2f\t\t %6.2f\n', P07/1e5, T07_a);
fprintf('8\t\t %5.2f\t\t %6.2f\n', P08/1e5, T08_a);

fprintf('HELIUM (Pressure Ratio: %.2f | Efficiency: %.2f%% | Net Work: %.2f kJ/m^3)',rc_opt_he, eta_th_he(
    idx_opt_he), W_net_he(idx_opt_he));
fprintf('Point\t Pressure (bar)\t Temperature (K)\n');
fprintf('1\t\t %5.2f\t\t %6.2f\n', P01/1e5, T01);
fprintf('2\t\t %5.2f\t\t %6.2f\n', P02_h/1e5, T02_h);
fprintf('3\t\t %5.2f\t\t %6.2f\n', P03_h/1e5, T03_h);
fprintf('4\t\t %5.2f\t\t %6.2f\n', P04_h/1e5, T04_h);
fprintf('5\t\t %5.2f\t\t %6.2f\n', P05_h/1e5, T05_h);
fprintf('6\t\t %5.2f\t\t %6.2f\n', P06_h/1e5, T06);
fprintf('7\t\t %5.2f\t\t %6.2f\n', P07/1e5, T07_h);
fprintf('8\t\t %5.2f\t\t %6.2f\n', P08/1e5, T08_h);
%% Based on Max Net Work


[~, idx_opt_W_air] = max(W_net_air);
rc_opt_W_air = rc_range(idx_opt_W_air);
[~, idx_opt_W_he] = max(W_net_he);
rc_opt_W_he = rc_range(idx_opt_W_he);

P02_a_W = P01 * sqrt(rc_opt_W_air);
T02_a_W = T01 * (P02_a_W/P01)^((gamma_air-1)/(gamma_air*eta_c));
P03_a_W = P02_a_W * 0.99;
P04_a_W = P01 * rc_opt_W_air;
T04_a_W = T03_a * (P04_a_W/P03_a_W)^((gamma_air-1)/(gamma_air*eta_c));
P05_a_W = P04_a_W * 0.975;
P06_a_W = P05_a_W * 0.97;
T07_a_W = T06 * (P07/P06_a_W)^(((gamma_air-1)*eta_t)/gamma_air);
T05_a_W = T04_a_W + epsilon * (T07_a_W - T04_a_W);
T08_a_W = T07_a_W - (T05_a_W - T04_a_W);

P02_h_W = P01 * sqrt(rc_opt_W_he);
T02_h_W = T01 * (P02_h_W/P01)^((gamma_he-1)/(gamma_he*eta_c));
P03_h_W = P02_h_W * 0.99;
P04_h_W = P01 * rc_opt_W_he;
T04_h_W = T03_h * (P04_h_W/P03_h_W)^((gamma_he-1)/(gamma_he*eta_c));
P05_h_W = P04_h_W * 0.975;
P06_h_W = P05_h_W * 0.97;
T07_h_W = T06 * (P07/P06_h_W)^(((gamma_he-1)*eta_t)/gamma_he);
T05_h_W = T04_h_W + epsilon * (T07_h_W - T04_h_W);
T08_h_W = T07_h_W - (T05_h_W - T04_h_W);

fprintf('Optimal point based on Max Net Work');
fprintf('AIR (Pressure Ratio for Max Work: %.2f | Efficiency: %.2f%% | Net Work: %.2f kJ/m^3)',rc_opt_W_air
    , eta_th_air(idx_opt_W_air), W_net_air(idx_opt_W_air));
fprintf('Point\t Pressure (bar)\t Temperature (K)\n');
fprintf('1\t\t %5.2f\t\t %6.2f\n', P01/1e5, T01);
fprintf('2\t\t %5.2f\t\t %6.2f\n', P02_a_W/1e5, T02_a_W);
fprintf('3\t\t %5.2f\t\t %6.2f\n', P03_a_W/1e5, T03_a);
fprintf('4\t\t %5.2f\t\t %6.2f\n', P04_a_W/1e5, T04_a_W);
fprintf('5\t\t %5.2f\t\t %6.2f\n', P05_a_W/1e5, T05_a_W);
fprintf('6\t\t %5.2f\t\t %6.2f\n', P06_a_W/1e5, T06);
fprintf('7\t\t %5.2f\t\t %6.2f\n', P07/1e5, T07_a_W);
fprintf('8\t\t %5.2f\t\t %6.2f\n', P08/1e5, T08_a_W);

fprintf('HELIUM (Pressure Ratio for Max Work: %.2f | Efficiency: %.2f%% | Net Work: %.2f kJ/m^3)',
    rc_opt_W_he, eta_th_he(idx_opt_W_he), W_net_he(idx_opt_W_he));
fprintf('Point\t Pressure (bar)\t Temperature (K)\n');
fprintf('1\t\t %5.2f\t\t %6.2f\n', P01/1e5, T01);
fprintf('2\t\t %5.2f\t\t %6.2f\n', P02_h_W/1e5, T02_h_W);
fprintf('3\t\t %5.2f\t\t %6.2f\n', P03_h_W/1e5, T03_h);
fprintf('4\t\t %5.2f\t\t %6.2f\n', P04_h_W/1e5, T04_h_W);
fprintf('5\t\t %5.2f\t\t %6.2f\n', P05_h_W/1e5, T05_h_W);
fprintf('6\t\t %5.2f\t\t %6.2f\n', P06_h_W/1e5, T06);
fprintf('7\t\t %5.2f\t\t %6.2f\n', P07/1e5, T07_h_W);
```

```
fprintf('8\t\t %5.2f\t\t %6.2f\n', P08/1e5, T08_h_W);
%% Utopia Point Optimization


W_norm_air = (W_net_air - min(W_net_air)) / (max(W_net_air) - min(W_net_air));
eta_norm_air = (eta_th_air - min(eta_th_air)) / (max(eta_th_air) - min(eta_th_air));
dist_air = sqrt((W_norm_air - 1).^2 + (eta_norm_air - 1).^2);
[~, idx_utopia_air] = min(dist_air);
rc_utopia_air = rc_range(idx_utopia_air);
W_norm_he = (W_net_he - min(W_net_he)) / (max(W_net_he) - min(W_net_he));
eta_norm_he = (eta_th_he - min(eta_th_he)) / (max(eta_th_he) - min(eta_th_he));
dist_he = sqrt((W_norm_he - 1).^2 + (eta_norm_he - 1).^2);
[~, idx_utopia_he] = min(dist_he);
rc_utopia_he = rc_range(idx_utopia_he);

figure('Units', 'normalized', 'Position', [0.1, 0.2, 0.8, 0.5]);
plot(W_net_air, eta_th_air, 'b-', 'LineWidth', 2); hold on;
plot(W_net_air(idx_utopia_air), eta_th_air(idx_utopia_air), 'ro', 'MarkerSize', 8, 'MarkerFaceColor', 'r');
grid on; box on;
xlabel('Specific Net Work (kJ/m^3)', 'FontSize', 11, 'FontWeight', 'bold');
ylabel('Thermal Efficiency (%)', 'FontSize', 11, 'FontWeight', 'bold');
title('Air: Efficiency vs. Net Work', 'FontSize', 13);
legend('Operating Curve', 'Ideal Operating Point', 'Location', 'best');

figure('Units', 'normalized', 'Position', [0.1, 0.2, 0.8, 0.5]);
plot(W_net_he, eta_th_he, 'r-', 'LineWidth', 2); hold on;
plot(W_net_he(idx_utopia_he), eta_th_he(idx_utopia_he), 'bo', 'MarkerSize', 8, 'MarkerFaceColor', 'b');
grid on; box on;
xlabel('Specific Net Work (kJ/m^3)', 'FontSize', 11, 'FontWeight', 'bold');
ylabel('Thermal Efficiency (%)', 'FontSize', 11, 'FontWeight', 'bold');
title('Helium: Efficiency vs. Net Work', 'FontSize', 13);
legend('Operating Curve', 'Ideal Operating Point', 'Location', 'best');

P02_a_U = P01 * sqrt(rc_utopia_air);
T02_a_U = T01 * (P02_a_U/P01)^((gamma_air-1)/(gamma_air*eta_c));
P03_a_U = P02_a_U * 0.99;
P04_a_U = P01 * rc_utopia_air;
T04_a_U = T03 * (P04_a_U/P03_a_U)^((gamma_air-1)/(gamma_air*eta_c));
P05_a_U = P04_a_U * 0.975;
P06_a_U = P05_a_U * 0.97;
T07_a_U = T06 * (P07/P06_a_U)^(((gamma_air-1)*eta_t)/gamma_air);
T05_a_U = T04_a_U + epsilon * (T07_a_U - T04_a_U);
T08_a_U = T07_a_U - (T05_a_U - T04_a_U);

P02_h_U = P01 * sqrt(rc_utopia_he);
T02_h_U = T01 * (P02_h_U/P01)^((gamma_he-1)/(gamma_he*eta_c));
P03_h_U = P02_h_U * 0.99;
P04_h_U = P01 * rc_utopia_he;
T04_h_U = T03 * (P04_h_U/P03_h_U)^((gamma_he-1)/(gamma_he*eta_c));
P05_h_U = P04_h_U * 0.975;
P06_h_U = P05_h_U * 0.97;
T07_h_U = T06 * (P07/P06_h_U)^(((gamma_he-1)*eta_t)/gamma_he);
T05_h_U = T04_h_U + epsilon * (T07_h_U - T04_h_U);
T08_h_U = T07_h_U - (T05_h_U - T04_h_U);

fprintf('Utopia Point Optimization ');
fprintf('AIR (Trade-off Pressure Ratio: %.2f | Efficiency: %.2f%% | Net Work: %.2f kJ/m^3)',rc_utopia_air,
    eta_th_air(idx_utopia_air), W_net_air(idx_utopia_air));
fprintf('Point\t Pressure (bar)\t Temperature (K)');
fprintf('1\t\t %5.2f\t\t %6.2f\n', P01/1e5, T01);
fprintf('2\t\t %5.2f\t\t %6.2f\n', P02_a_U/1e5, T02_a_U);
fprintf('3\t\t %5.2f\t\t %6.2f\n', P03_a_U/1e5, T03);
fprintf('4\t\t %5.2f\t\t %6.2f\n', P04_a_U/1e5, T04_a_U);
fprintf('5\t\t %5.2f\t\t %6.2f\n', P05_a_U/1e5, T05_a_U);
fprintf('6\t\t %5.2f\t\t %6.2f\n', P06_a_U/1e5, T06);
fprintf('7\t\t %5.2f\t\t %6.2f\n', P07/1e5, T07_a_U);
fprintf('8\t\t %5.2f\t\t %6.2f\n', P08/1e5, T08_a_U);
fprintf(' HELIUM (Trade-off Pressure Ratio: %.2f | Efficiency: %.2f%% | Net Work: %.2f kJ/m^3)',
    rc_utopia_he, eta_th_he(idx_utopia_he), W_net_he(idx_utopia_he));
fprintf('Point\t Pressure (bar)\t Temperature (K)\n');
fprintf('1\t\t %5.2f\t\t %6.2f\n', P01/1e5, T01);
fprintf('2\t\t %5.2f\t\t %6.2f\n', P02_h_U/1e5, T02_h_U);
fprintf('3\t\t %5.2f\t\t %6.2f\n', P03_h_U/1e5, T03);
```

```
fprintf('4\t\t %5.2f\t\t %6.2f\n', P04_h_U/1e5, T04_h_U);
fprintf('5\t\t %5.2f\t\t %6.2f\n', P05_h_U/1e5, T05_h_U);
fprintf('6\t\t %5.2f\t\t %6.2f\n', P06_h_U/1e5, T06);
fprintf('7\t\t %5.2f\t\t %6.2f\n', P07/1e5, T07_h_U);
fprintf('8\t\t %5.2f\t\t %6.2f\n', P08/1e5, T08_h_U);

%% Heat Exchanger effect on efficiency (task 5)


epsilon_array = [0.7, 0.8, 0.9];
num_eps = length(epsilon_array);
eta_th_air_eps = zeros(num_eps, N);
eta_th_he_eps  = zeros(num_eps, N);

for k = 1:num_eps
    eps_current = epsilon_array(k);

    for i = 1:N
        rc = rc_range(i);

        P02_a = P01 * sqrt(rc);
        P03_a = P02_a * 0.99;
        P04_a = P01 * rc;
        P05_a = P04_a * 0.975;
        P06_a = P05_a * 0.97;
        T02_a = T01 * (P02_a/P01)^((gamma_air-1)/(gamma_air*eta_c));
        T04_a = T03 * (P04_a/P03_a)^((gamma_air-1)/(gamma_air*eta_c));
        T07_a = T06 * (P07/P06_a)^(((gamma_air-1)*eta_t)/gamma_air);
        T05_a = T04_a + eps_current * (T07_a - T04_a);

        wc1_a = cp_air * (T02_a - T01);
        wc2_a = cp_air * (T04_a - T03);
        wt_a  = cp_air * (T06 - T07_a);
        wnet_a = wt_a - (wc1_a + wc2_a);
        qin_a  = cp_air * (T06 - T05_a);
        eta_th_air_eps(k, i) = (wnet_a / qin_a) * 100;

        P02_h = P01 * sqrt(rc);
        P03_h = P02_h * 0.99;
        P04_h = P01 * rc;
        P05_h = P04_h * 0.975;
        P06_h = P05_h * 0.97;
        T02_h = T01 * (P02_h/P01)^((gamma_he-1)/(gamma_he*eta_c));
        T04_h = T03 * (P04_h/P03_h)^((gamma_he-1)/(gamma_he*eta_c));
        T07_h = T06 * (P07/P06_h)^(((gamma_he-1)*eta_t)/gamma_he);
        T05_h = T04_h + eps_current * (T07_h - T04_h);

        wc1_h = cp_he * (T02_h - T01);
        wc2_h = cp_he * (T04_h - T03);
        wt_h  = cp_he * (T06 - T07_h);
        wnet_h = wt_h - (wc1_h + wc2_h);
        qin_h  = cp_he * (T06 - T05_h);
        eta_th_he_eps(k, i) = (wnet_h / qin_h) * 100;
    end
end

figure('Units', 'normalized', 'Position', [0.1, 0.2, 0.8, 0.5]);
plot(rc_range, eta_th_air_eps(1,:), 'b-',  'LineWidth', 2); hold on;
plot(rc_range, eta_th_air_eps(2,:), 'b--', 'LineWidth', 2);
plot(rc_range, eta_th_air_eps(3,:), 'b:',  'LineWidth', 2.5);
grid on; box on;
xlabel('Total Pressure Ratio (P_{04}/P_{01})', 'FontSize', 11, 'FontWeight', 'bold');
ylabel('Thermal Efficiency (%)', 'FontSize', 11, 'FontWeight', 'bold');
title('Air: Effect of Effectiveness (\epsilon)', 'FontSize', 13);
legend('\epsilon = 0.7', '\epsilon = 0.8', '\epsilon = 0.9', 'Location', 'best');

figure('Units', 'normalized', 'Position', [0.1, 0.2, 0.8, 0.5]);
plot(rc_range, eta_th_he_eps(1,:), 'r-',  'LineWidth', 2); hold on;
plot(rc_range, eta_th_he_eps(2,:), 'r--', 'LineWidth', 2);
plot(rc_range, eta_th_he_eps(3,:), 'r:',  'LineWidth', 2.5);
grid on; box on;
xlabel('Total Pressure Ratio (P_{04}/P_{01})', 'FontSize', 11, 'FontWeight', 'bold');
ylabel('Thermal Efficiency (%)', 'FontSize', 11, 'FontWeight', 'bold');
```

```
342 title('Helium: Effect of Effectiveness (\epsilon)', 'FontSize', 13);
343 legend('\epsilon = 0.7', '\epsilon = 0.8', '\epsilon = 0.9', 'Location', 'best');
344
345 %% task 6 - Heat exchanger properties and cost comparison
346
347 %%Cohen section 2.6:
348 %%Nu = 0.023 Re^0.8 Pr^0.4
349 %%h_He / h_air ≈ 1.86
350 eps_opt = 0.9;
351 NTU_opt = eps_opt / (1 - eps_opt);
352 fprintf('NTU = epsilon/(1-epsilon) = %.3f\n', NTU_opt);
353
354 rc6_air = rc_utopia_air;
355 rc6_he  = rc_utopia_he;
356
357 P02_a = P01 * sqrt(rc6_air);
358 T02_a = T01 * (P02_a/P01)^((gamma_air-1)/(gamma_air*eta_c));
359 P03_a = P02_a * 0.99;
360 T03_a = T03;
361 P04_a = P01 * rc6_air;
362 T04_a = T03_a * (P04_a/P03_a)^((gamma_air-1)/(gamma_air*eta_c));
363 P05_a = P04_a * 0.975;
364 P06_a = P05_a * 0.97;
365 P08_a = P08;
366 P07_a = P07;
367 T07_a = T06 * (P07_a/P06_a)^(((gamma_air-1)*eta_t)/gamma_air);
368 T05_a = T04_a + eps_opt * (T07_a - T04_a);
369 T08_a = T07_a - (T05_a - T04_a);
370
371 Tmean_HX_air = mean([T04_a, T05_a, T07_a, T08_a]);
372
373 P02_h = P01 * sqrt(rc6_he);
374 T02_h = T01 * (P02_h/P01)^((gamma_he-1)/(gamma_he*eta_c));
375 P03_h = P02_h * 0.99;
376 T03_h = T03;
377 P04_h = P01 * rc6_he;
378 T04_h = T03_h * (P04_h/P03_h)^((gamma_he-1)/(gamma_he*eta_c));
379 P05_h = P04_h * 0.975;
380 P06_h = P05_h * 0.97;
381 P08_h = P08;
382 P07_h = P07;
383 T07_h = T06 * (P07_h/P06_h)^(((gamma_he-1)*eta_t)/gamma_he);
384 T05_h = T04_h + eps_opt * (T07_h - T04_h);
385 T08_h = T07_h - (T05_h - T04_h);
386
387 Tmean_HX_he = mean([T04_h, T05_h, T07_h, T08_h]);
388
389 fprintf('Air HX mean temperature = %.3f K\n', Tmean_HX_air);
390 fprintf('Helium HX mean temperature = %.3f K\n', Tmean_HX_he);
391
392 %%assume
393 Cdot_basis = 1.0;
394 U_air = 0.100;
395
396 %% @ mean HX temperature
397 T_air_table = [300 350 400 450 500 550 600 650 700 750 800 850 900 950 1000 1100 1200 1300 1400 1500 1600];
398 mu_air_table = [184.6 208.2 230.1 250.7 270.1 288.4 305.8 322.5 338.8 354.6 369.8 384.3 398.1 411.3 424.4
        449.0 473.0 496.0 530.0 557.0 584.0];
399 k_air_table = [26.3 30.0 33.8 37.3 40.7 43.9 46.9 49.7 52.4 54.9 57.3 59.6 62.0 64.3 66.7 71.5 76.3 82.0
        91.0 100.0 106.0];
400
401 T_he_table = [300 350 400 450 500 600 650 700 750 800 900 1000];
402 mu_he_table = [199 221 243 263 283 320 332 350 364 382 414 446];
403 k_he_table = [152 170 187 204 220 252 264 278 291 304 330 354];
404
405 mu_air_HX = interp1(T_air_table, mu_air_table, Tmean_HX_air, 'linear', 'extrap');
406 k_air_HX  = interp1(T_air_table, k_air_table, Tmean_HX_air, 'linear', 'extrap');
407
408 mu_he_HX = interp1(T_he_table, mu_he_table, Tmean_HX_he, 'linear', 'extrap');
409 k_he_HX  = interp1(T_he_table, k_he_table, Tmean_HX_he, 'linear', 'extrap');
410
411 rho_ratio = R_air / R_he;
412 mu_ratio  = mu_he_HX / mu_air_HX;
```

```
k_ratio   = k_he_HX / k_air_HX;

C_ratio = (1 / (rho_ratio^0.75 * mu_ratio^0.25))^(1/1.75);
Re_ratio = rho_ratio * C_ratio / mu_ratio;

h_ratio_calc = (Re_ratio^0.8) * k_ratio;
h_ratio_cohen = 1.86;

if abs(h_ratio_calc - h_ratio_cohen)/h_ratio_cohen > 0.10
    h_ratio = h_ratio_cohen;
else
    h_ratio = h_ratio_calc;
end

U_he = h_ratio * U_air;

fprintf('rho_He/rho_air at HX = %.3f\n', rho_ratio);
fprintf('mu_He/mu_air at HX = %.3f\n', mu_ratio);
fprintf('k_He/k_air at HX = %.3f\n', k_ratio);
fprintf('Calculated h_He/h_air = %.3f\n', h_ratio_calc);
fprintf('Used h_He/h_air = %.3f\n', h_ratio);
fprintf('Calculated U_He/U_air = %.3f\n', U_he/U_air);

A_air = NTU_opt * Cdot_basis / U_air;
A_he  = NTU_opt * Cdot_basis / U_he;

area_ratio = A_he / A_air;
area_reduction_percent = (1 - area_ratio) * 100;

fprintf('A_air = %.3f m^2\n', A_air);
fprintf('A_He  = %.3f m^2\n', A_he);
fprintf('A_He/A_air = %.3f\n', area_ratio);
fprintf('Helium heat exchanger area reduction = %.2f %%\n', area_reduction_percent);

alpha = 15;

M_air = alpha * A_air;
M_he  = alpha * A_he;

mass_reduction_percent = (1 - M_he / M_air) * 100;

fprintf('M_air = %.3f kg\n', M_air);
fprintf('M_He  = %.3f kg\n', M_he);
fprintf('Mass reduction = %.2f %%\n', mass_reduction_percent);

C0 = @(A) 10.^(4.3247 - 0.3030 .* log10(A) + ...
               0.1634 .* (log10(A)).^2);

Cost_air = C0(A_air);
Cost_he  = C0(A_he);

cost_reduction_percent = (1 - Cost_he / Cost_air) * 100;

fprintf('Cost_air = %.2f USD\n', Cost_air);
fprintf('Cost_He  = %.2f USD\n', Cost_he);
fprintf('Cost reduction = %.2f %%\n', cost_reduction_percent);
```

Chapter B

Nomenclature

Table B.1: List of symbols used in this report

Symbol	Unit	Description
A	m^2	Heat-transfer area
C^0	USD	Estimated heat-exchanger base cost
C_{min}	kW/K	Minimum heat-capacity rate
c_p	kJ/kg.K	Specific heat at constant pressure
h	kW/m^2K	Convective heat-transfer coefficient
M	kg	Heat-exchanger mass
NTU	–	Number of transfer units
P	bar or Pa	Pressure
q_{in}	kJ/kg	External heat input per unit mass
R	J/kg.K	Gas constant
r_c	–	Total compressor pressure ratio
Re	–	Reynolds number
T	K	Temperature
U	kW/m^2K	Overall heat-transfer coefficient
w_c	kJ/kg	Compressor work per unit mass
w_t	kJ/kg	Turbine work per unit mass
w_{net}	kJ/kg	Net work per unit mass
$W_{net,vol}$	kJ/m^3	Volumetric specific net work
α	kg/m^2	Heat-exchanger mass-area coefficient
ϵ	–	Recuperator effectiveness
η_c	–	Compressor polytropic efficiency
η_t	–	Turbine polytropic efficiency
η_{th}	%	Thermal efficiency
γ	–	Heat-capacity ratio
ρ	kg/m^3	Density
μ	Pa.s	Dynamic viscosity
k	W/m.K	Thermal conductivity

References

[1] H.I.H. Saravanamuttoo, G.F.C. Rogers, H. Cohen, P.V. Straznicky, and A.C. Nix, "Gas Turbine Theory", 7th Edition, Pearson, 2017.

[2] C.F. McDonald, "Helium turbomachinery operating experience from gas turbine power plants and test facilities", Applied Thermal Engineering, Vol. 44, pp. 108–142, 2012.

[3] C. Keller, "Forty years of experience on closed-cycle gas turbines", Progress in Nuclear Energy, Vol. 2, No. 1, pp. 77–99, 1978.

[4] V. Mrzljak, I. Poljak, J. Prpic-Orsic, and M. Jelic, "Thermodynamic Analysis and Improvement Potential of Helium Closed-Cycle Gas Turbine Power Plant at Four Loads", Energies, Vol. 16, No. 15, 2023.

[5] Z. Zhang, Z. Wu, D. Wang, Y. Xu, Y. Sun, F. Li, and Y. Dong, "The Shandong Shidao Bay 200 MWe High-Temperature Gas-Cooled Reactor Pebble-Bed Module Demonstration Power Plant", Engineering, Vol. 2, No. 1, pp. 112–118, 2016.

[6] World Nuclear News, "China's demonstration HTR-PM enters commercial operation", 2023.

[7] Y. Ahn, S.J. Bae, M. Kim, S.K. Cho, S. Baik, J.I. Lee, and J.E. Cha, "Review of supercritical CO2 power cycle technology and current status of research and development", Nuclear Engineering and Technology, Vol. 47, No. 6, pp. 647–661, 2015.

[8] GTI Energy, "STEP Demo Pilot Achieves Phase 1 Testing Milestone", 2024.

[9] F.P. Incropera, D.P. DeWitt, T.L. Bergman, and A.S. Lavine, "Fundamentals of Heat and Mass Transfer", 6th Edition, Wiley, 2007.